Model Alpha M14 Benchmark Design Methods Study

HELIX FORGE LAB / p04-full-006756 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded benchmark design evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000149 (research/programs/model-alpha/experiments/results/2026-07-13/model-alpha-m14/reproducibility/specifications/model-alpha-m14-reprodu

Slice	Records	Precision	Recall	F1
slice-01	160	0.877	0.878	0.877
slice-02	197	0.894	0.854	0.874
slice-03	234	0.876	0.867	0.871
slice-04	223	0.893	0.880	0.886
slice-05	236	0.875	0.856	0.865